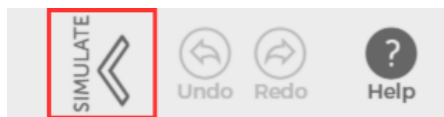


Module 3.4: Impact of Carbon Dioxide Emissions on the Ocean's pH and Food Web

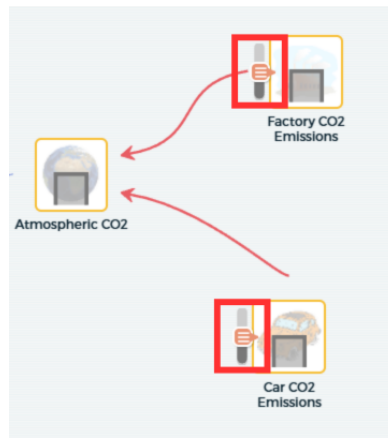
Day 2: Sage Modeler Instructions

Follow these steps with your team and use SageModeler to explore how the ocean's pH and foodweb are impacted by carbon dioxide emissions.

1. Visit the [SageModeler](#) website. The first scenario you will explore is how seawater is impacted by carbon dioxide emissions.
2. As a group, take a moment to make observations of the model. In particular, view the connections to the icons and their potential relationships. Answer the following questions:
 - How does carbon dioxide get introduced into the Earth's system in the model?
3. After you've made your observations, take a moment to make a prediction with your group on how you think increasing carbon dioxide emissions might impact the ocean.
 - How might increases in these sources of carbon dioxide impact the Earth's system?
4. Once you have made your prediction, it is time to test it. To test your prediction, click on the "**Simulate**" menu in the upper right-hand corner of the platform. You'll notice that your model in the workspace changes.



5. You'll notice that sliding bars appear next to **Factory CO₂ Emissions** and **Car CO₂ Emissions**. These are variables that you can modify. The gray bars signify the measurement of its respective variable.



6. Take a moment to manipulate the simulation to make the ocean's pH more basic (higher ocean pH).
 - What do you notice about the system when the ocean's pH is more basic?
 - How do basic conditions in the ocean's pH impact the organisms in the food web?

7. Now, make the ocean's pH more acidic by adjusting your model (lower ocean pH).
 - What do you notice about the system when the ocean's pH is more acidic?
 - How does making the conditions more acidic in the ocean's pH impact the organisms compared to basic conditions?

8. What differences do you notice between increasing and decreasing the amount of carbon dioxide and the impacts it has on the living organisms in this model?